Dynamic Social Optimum: Diamond Network

COMP3821

1 Definition and reduction to path assignments

The simulator uses linear edge costs $c_e(f) = a_e + b_e f$ and the total social cost

$$SC = \sum_t \sum_{e \in E} f_e(t)\, c_e\big(f_e(t)\big),$$

where $f_e(t)$ is the number of agents that traverse e during timestep t. The *dynamic social optimum* is the lowest value of this expression over every feasible way of routing the agents:

$$\boxed{SC_{\text{opt}} = \min_{\text{feasible routing decisions}} \sum_t \sum_{e \in E} f_e(t)\, c_e\big(f_e(t)\big)}$$

Feasible routings. The movement rules of the specification (Section 3.1) and the simulator fix what "feasible" means:

1. all N agents are released at $t = 0$ (none of these scenarios uses cascades);
2. at every timestep each unfinished agent traverses exactly one edge, so no agent can wait;
3. with `forward_only` (the default), an agent never moves to a node with smaller hop distance from its origin and never revisits a node.

Lemma 1 (Routing = path assignment)**.** *Let $\mathcal{P}_{od}$ be the simple paths from o to d that obey rule 3. Every feasible routing gives each agent a path $P_i \in \mathcal{P}_{o_i d_i}$, and agent i traverses the k-th edge of P_i at timestep $t = k - 1$. Every assignment of such paths can be realised. Therefore*

$$SC = \sum_{e \in E} \sum_{t \geq 0} X_{e,t}\,\big(a_e + b_e X_{e,t}\big), \qquad X_{e,t} = \#\{\, i : e \text{ is the } (t+1)\text{-th edge of } P_i \,\},$$

and SC_{opt} is the minimum of this over all path assignments.

Proof. Rule 2 means an agent's position at time t is the end of the t-th edge it has taken. Rule 3 means the edges it takes form a simple path in the allowed subgraph. So $f_e(t) = X_{e,t}$. Conversely, a central planner can tell each agent which next edge to commit to at each step. The outcome depends only on the paths, so every path assignment is achievable. □

So the dynamic problem is a *static* problem on the *time-expanded* resources (e, t). Agents on the same edge only slow each other down if they cross it in the same step.

Lemma 2 (When the dynamic and static optima coincide)**.** *Suppose every edge e sits at the same position $k(e)$ on every feasible path that contains it. Then $X_{e,t} = x_e\,\mathbf{1}[t = k(e) - 1]$, where x_e is the total number of agents using e. So $SC = \sum_e x_e\, c_e(x_e)$, which is the classical static social-optimum objective (Beckmann; Roughgarden) restricted to integer flows.*

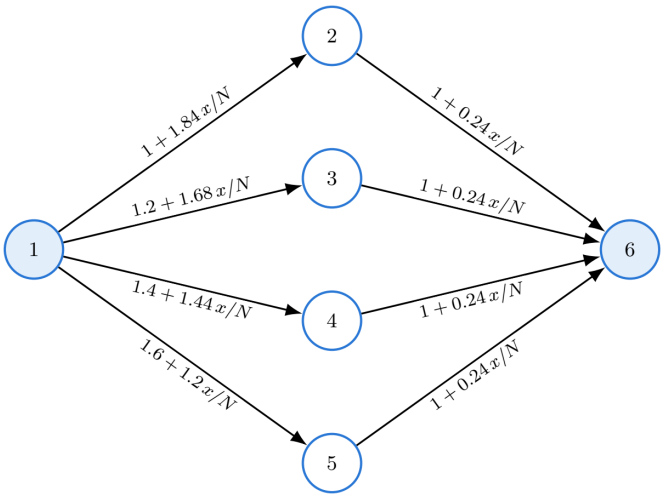
Integer versus continuous. Agents are indivisible, so SC_{opt} is an integer program. Relaxing the path counts to non-negative reals gives a convex quadratic program. Its minimum is a lower bound on SC_{opt} and equals the limit of SC_{opt}/N for large N. Both are reported below. Because c_e is linear, the cost one extra agent adds to resource (e,t) is

$$\Delta^{+}_{e,t} = (X+1)\,c_e(X+1) - X\,c_e(X) = a_e + b_e(2X+1), \qquad X = X_{e,t},$$

and in the continuous relaxation the marginal cost is $\frac{d}{dX}\big[X c_e(X)\big] = a_e + 2b_e X$.

Scaling with N. Every small scenario sets $b_e = \beta_e/N$ with β_e fixed, so the per-agent optimum SC_{opt}/N is almost the same for $N = 20, 50, 80$.

2 Network



All agents travel $1 \to 6$ along one of four disjoint two-edge routes $R_k = 1 \to k \to 6$, $k = 2, \ldots, 5$.

Route	a_{1k}	$b_{1k}N$	a_{k6}	$b_{k6}N$	$A_k = a_{1k} + a_{k6}$	$\beta_k = (b_{1k} + b_{k6})N$
R_2	1.0	1.84	1.0	0.24	2.0	2.08
R_3	1.2	1.68	1.0	0.24	2.2	1.92
R_4	1.4	1.44	1.0	0.24	2.4	1.68
R_5	1.6	1.20	1.0	0.24	2.6	1.44

The scenarios use $N = 20$ (`diamond_N20`), 50 (`diamond_N50`) and 80 (`diamond`). For example, at $N = 80$, $b_{12} = 0.023$ and $b_{26} = 0.003$.

3 Derivation

Timing. Every route has two edges. Edge $1 \to k$ is always crossed at $t = 0$ and $k \to 6$ at $t = 1$. Lemma 2 applies, so the dynamic and static optima coincide.

Objective. The routes share no edges. With n_k agents on R_k, both of its edges carry n_k agents, so

$$SC(n) = \sum_{k=2}^{5} \Big[n_k(a_{1k} + b_{1k}n_k) + n_k(a_{k6} + b_{k6}n_k) \Big] = \sum_{k=2}^{5} \Big[A_k n_k + \frac{\beta_k}{N} n_k^2 \Big], \qquad \sum_k n_k = N.$$

This is a *separable* convex resource-allocation problem.

Continuous optimum (KKT). With fractions $x_k = n_k/N$, $SC/N = \sum_k (A_k x_k + \beta_k x_k^2)$. At the optimum every used route has the same marginal cost λ, and unused routes have marginal cost at least λ:

$$A_k + 2\beta_k x_k = \lambda \quad (x_k > 0) \qquad \Longrightarrow \qquad x_k = \frac{\lambda - A_k}{2\beta_k}.$$

Assume all four routes are used (checked below). Then $\sum_k x_k = 1$ gives

$$\lambda = \frac{1 + \sum_k A_k/(2\beta_k)}{\sum_k 1/(2\beta_k)} = \frac{96203}{30025} \approx 3.20410.$$

Route	x_k (exact)	x_k	Cost per agent $A_k + \beta_k x_k$	Marginal $A_k + 2\beta_k x_k$
R_2	2781/9608	0.28945	2.60205	3.20410
R_3	7537/28824	0.26148	2.70205	3.20410
R_4	3449/14412	0.23931	2.80205	3.20410
R_5	3023/14412	0.20976	2.90205	3.20410

Since $\lambda > \max_k A_k = 2.6$, every $x_k > 0$, so the assumption holds. The objective is strictly convex, so this point is the unique continuous optimum:

$$\frac{SC^*}{N} = \sum_k \left(A_k x_k + \beta_k x_k^2\right) = \frac{1315809}{480400} \approx 2.738986.$$

The routes have equal *marginal* costs, but each agent's own cost differs by exactly 0.1 per route. Cheap routes are deliberately loaded more, so their users pay less.

Integer optimum (exact). The j-th agent placed on R_k raises the cost by

$$\mu_{k,j} = \left[A_k j + \tfrac{\beta_k}{N} j^2\right] - \left[A_k (j-1) + \tfrac{\beta_k}{N}(j-1)^2\right] = A_k + \frac{\beta_k}{N}(2j-1),$$

which increases in j. Hence $SC(n) = \sum_k \sum_{j \le n_k} \mu_{k,j}$ is a sum of N marginals, and every route uses a prefix $\mu_{k,1}, \ldots, \mu_{k,n_k}$. Any N marginals cost at least the N smallest in the list $\{\mu_{k,j}\}$. Because each route's marginals increase, the N smallest automatically form a prefix on each route. So taking the N smallest marginals (equivalently, adding agents one at a time to the route with the cheapest next marginal) is exactly optimal. The N-th and $(N+1)$-th smallest marginals are distinct in every case, so the optimum is unique.

4 Result

N	n_2	n_3	n_4	n_5	N-th / $(N{+}1)$-th marginal	SC_{opt}	SC_{opt}/N	Cont. bound/N
20	6	5	5	4	3.1560 / 3.2480	54.7960	2.73980	2.73899
50	14	13	12	11	3.2048 / 3.2064	136.9664	2.73933	2.73899
80	23	21	19	17	3.1940 / 3.2190	219.1210	2.73901	2.73899

$$\boxed{SC_{\text{opt}}/N \to \tfrac{1315809}{480400} \approx 2.73899 \quad (x \approx (0.289,\, 0.261,\, 0.239,\, 0.210))}$$

Verification. The values were checked three independent ways, and all three agree exactly: an exact convex min-cost-flow solver on the time-expanded network, the continuous relaxation, and single-agent local search.